Replication Instructions – Sentiment Analysis of Amazon Reviews

This guide walks through how to fully replicate our project using Google Cloud Platform (GCP) and Dataproc. The full workflow can be run from a Jupyter notebook-enabled cluster using the provided notebooks in order.

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1. Prerequisites

- Google Cloud Platform account (with billing enabled)
- Kaggle account
- A Google Cloud Storage (GCS) bucket (e.g., `gs://your-bucket-name`)

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2. Setting Up Kaggle Credentials

1. Go to Kaggle → My Account → API
2. Click **"Create New API Token"**
3. This will download a `kaggle.json` file to your local machine
4. Upload that file to your Dataproc Jupyter notebook environment before running the first notebook

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3. GCP Cluster Setup (Dataproc)

Use the following configuration when creating a Dataproc cluster:

- **Cluster name**: `sentiment-cluster`
- **Region**: `us-central1`
- **Master node**: `n4-standard-4` (4 vCPU, 16 GB RAM)
- **Worker nodes**: 2 × `n4-standard-2` (2 vCPU, 8 GB RAM)
- **Enable Jupyter Notebook**
- **Important**: In the Networking tab, **uncheck "Internal IP only"** so the cluster can access the internet

Create or use an existing GCS bucket to store input data and output artifacts.

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4. Notebook Execution Order

All notebooks are provided inside the `/notebooks/` directory. Run them in the following order:

Step 1 – Download Dataset

Notebook: `0_download_dataset.ipynb`
- Uses Kaggle API to download Amazon reviews dataset
- Extracts selected categories and uploads `.tsv` files to GCS

Step 2 – Data Loading and Joining

Notebook: `1_data_loading_joining.ipynb`
- Loads TSVs from GCS and parses review text
- Adds helpful_ratio and review_length
- Combines all categories and saves cleaned data as Parquet

Step 3 – Initial EDA

Notebook: `2_initial_eda.ipynb`
- Plots distribution of star ratings and review length
- Shows data skew and correlation patterns

Step 4 – Data Cleaning

Notebook: `3_data_cleaning.ipynb`
- Filters short or invalid reviews
- Adds binary flags (has_votes, verified_purchase) and temporal fields

Step 5 – Baseline Sentiment Modeling

Notebook: `4_review_sentiment_modeling.ipynb`
- Trains logistic regression model using TF-IDF and metadata features
- Evaluates accuracy and F1 score

Step 6 – Hyperparameter Tuning and Final Model

Notebook: `5_hyperparameter_sentiment.ipynb`
- Adds sentiment score, positive/negative word ratios, and Word2Vec embeddings
- Performs model tuning (logistic + random forest)
- Outputs final metrics and saves model

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5. Output and Replication

Each notebook saves its outputs to GCS as Parquet, models, or charts.
The final model and Word2Vec vectors can also be saved and reused.

To replicate the project end-to-end: - Follow this notebook sequence - Set the GCS paths as needed in each notebook - Use Dataproc Spark kernels or PySpark via Jupyter

This setup will fully recreate the pipeline, from ingestion to sentiment classification.